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Sreevetha Institute of Medical And Technical Sciences
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CSA15 Cloud Computing and Big Data Analytics

CO1 AT2 - Digital Concept Map and Service Model Matrix

Individual Student Submission Template

Student and Faculty Details

Student Name	Karanam Reddy Vishnu Chaithanya	Register Number	192372057
Class / Section		Slot	B
Course Code	CSA-1512	Course Title	Cloud Computng and Big data Analytics
Capstone Title	SentinelX Cloud Threat Intelligence Using NLP and Serverless Risk Scoring		
Submission Date	13-07-2026		
Faculty Name	Dr.Rajesh Babu C	Employee ID	
Department	CSE-AI		

Assessment Brief

Course Outcome	CO1 - Explain the evolution, architecture, characteristics, service models, and emerging paradigms of cloud computing.
Assessment Type	Individual concept mapping, service-model responsibility analysis, and capstone-cloud interpretation.
Assessment Focus	Each map must show key nodes, link labels, relationship meaning, and a logical hierarchy. The response must connect cloud fundamentals to the student's own perception of the capstone title.
Submission Mode	Editable DOCX + exported diagram evidence + GitHub repository link submitted in Google Classroom.
Permitted Tools	draw.io / diagrams.net, Lucidchart, Miro, Canva, PowerPoint, Google Drawings, or equivalent diagramming tool.
Individual Rule	Every student must submit their own concept map, matrix, capstone-cloud mapping, and reflection. The capstone title may be shared by a team, but the assessment answer must not be common.
Total Marks	100 marks

Code of Conduct and Student Assurance

Academic Integrity Statement

This assessment must be completed individually. Students may discuss the capstone domain with their team, but the concept map, service model matrix, explanations, GitHub evidence, and reflection submitted here must be individually prepared.

Student Assurance	I confirm that this submission represents my own understanding of CO1 concepts and my individual interpretation of the selected capstone title.
Tool Use	I have used digital tools only for diagram preparation, organization, and presentation. I have not copied another student's diagram, matrix, or explanation.
Evidence	I have uploaded the required files to my GitHub repository and submitted the repository link / evidence link through Google Classroom.
Student Signature	
Date	13-17-2026

Procedure for Preparing the Digital Concept Map

Use the following procedure to prepare the diagram and upload evidence. Students who already know another diagramming tool may use it, but the exported evidence and source file must still be submitted.

1. Open <https://app.diagrams.net/> or the desktop version of draw.io / diagrams.net.
2. Choose Device, Google Drive, or OneDrive as the save location according to availability.
3. Create a blank diagram and name the file as CSA15_CO1_AT2_RegisterNumber_Concept_Map.drawio.
4. Place the capstone title at the center or as a clearly visible anchor node.
5. Add major cloud nodes: cloud definition, cloud evolution, cloud architecture, service models, deployment models, virtualization, web services, SOAP, REST, APIs, elasticity, scalability, resource pooling, measured service, and emerging paradigms.
6. Connect nodes with labeled arrows such as uses, depends on, enables, scales through, communicates through, belongs to, supports, provides, consumes, or controls.
7. Use colors meaningfully: one color for service models, one for cloud characteristics, one for API/web-service concepts, and one for capstone-specific elements.
8. Export the final diagram as PNG or PDF. Keep the editable .drawio source file also.
9. Paste the exported diagram image or screenshot in this DOCX under Task 1.
10. Upload the DOCX, exported PNG/PDF, and .drawio file to GitHub, then submit the GitHub link in Google Classroom.

GitHub Submission Procedure

Step	Required Action	Expected Evidence
1	Create or use your individual GitHub account.	GitHub profile link.
2	Create a repository named CSA15-CO-Assessments-RegisterNumber or use the repository instructed by faculty.	Repository URL.
3	Create a folder named CO1_AT2 inside the repository.	Folder visible in repository.
4	Upload this completed DOCX, exported diagram PNG/PDF, and editable .drawio source file.	At least three evidence files uploaded.
5	Add a short README.md explaining the capstone title and files submitted.	README visible in the AT2 folder.
6	Submit the GitHub folder link in Google Classroom.	GCR submission with repository/folder link.

Assessment Questions and Student Response Sections

Important

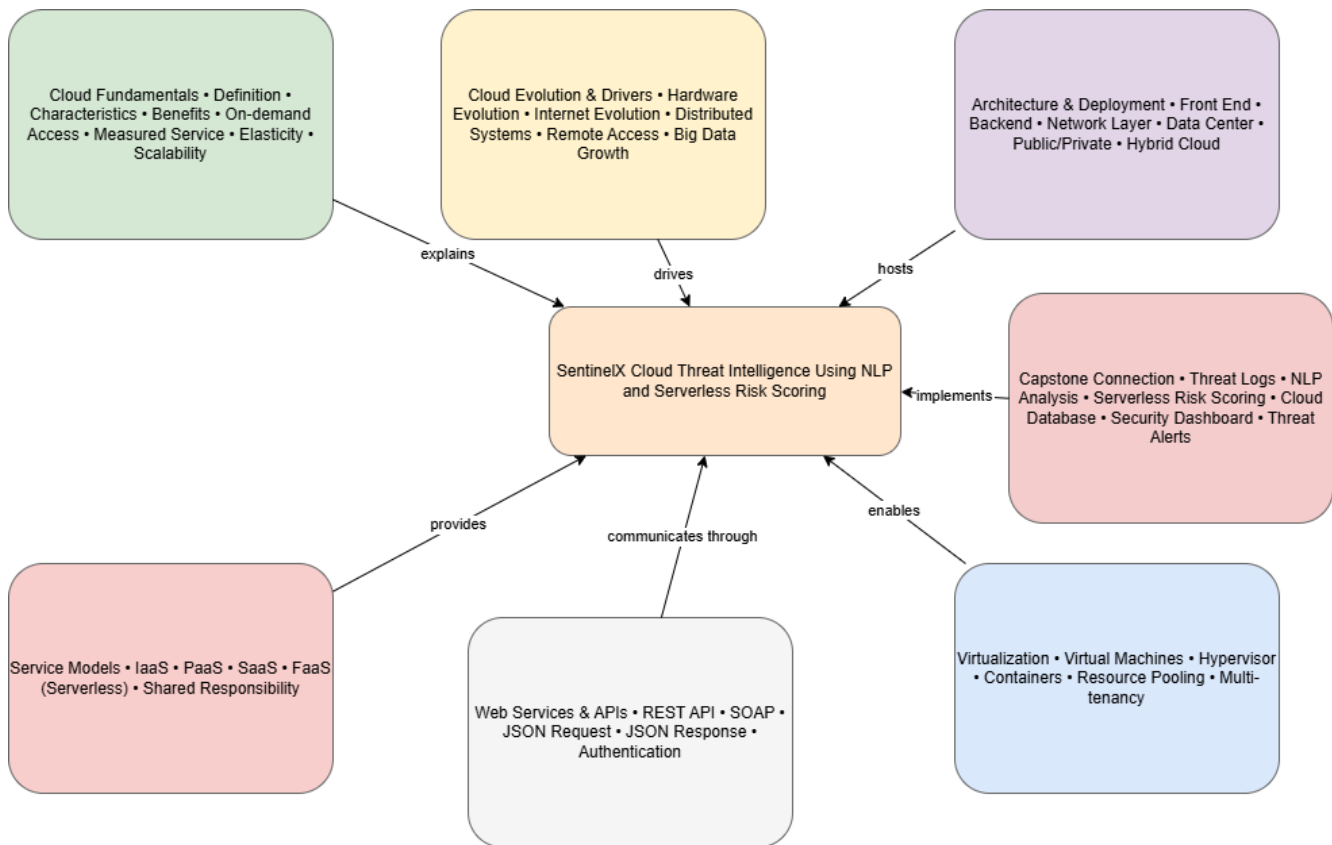
The capstone project may be developed by a team, but every response below must be written from the individual student's own perspective. Do not submit a common team answer.

Task 1 - Digital Cloud Concept Map

Create one integrated digital concept map for CO1. The map must show key nodes, link labels, relationship meaning, and a logical hierarchy. It must also connect the concepts to your selected capstone title from your individual perspective.

Mandatory Concept Group	Minimum Concepts / Nodes to Include
Cloud Fundamentals	Definition, characteristics, benefits, limitations, on-demand access, measured service.
Cloud Evolution and Drivers	Hardware evolution, internet software evolution, distributed systems, cost economics, mobility, remote access, big data growth.
Architecture and Deployment	Front-end, back-end, network layer, data center, management software, public/private/hybrid/community cloud.
Service Models and Responsibility	IaaS, PaaS, SaaS, XaaS, provider responsibility, user responsibility, shared responsibility.
Virtualization and Web/API Layer	Virtual machine, hypervisor, abstraction, resource pooling, multi-tenancy, web services, SOAP, REST, API request/response.
Capstone Connection	Your capstone users, app/module, cloud backend, database/storage, dashboard/analytics, scalability and elasticity idea.

Paste exported concept map image / screenshot here. Resize neatly within this white space.



Brief explanation of your concept map: explain the most important nodes, link labels, hierarchy, and capstone connection.

The concept map presents the fundamental concepts of cloud computing and explains how they are connected to the SentinelX Cloud Threat Intelligence Using NLP and Serverless Risk Scoring project. Cloud Fundamentals form the base of the system by providing on-demand services, scalability, elasticity, and measured service. Cloud Evolution highlights the technological developments that made cloud platforms practical for modern applications. The Architecture and Deployment section illustrates the interaction between the frontend, backend, network, and cloud infrastructure. The Service Models (IaaS, PaaS, SaaS, and FaaS/Serverless) define the responsibilities shared between the cloud provider and the development team. Virtualization enables efficient resource sharing through virtual machines, containers, and resource pooling. Web Services and REST APIs allow secure communication between the web application, NLP engine, serverless risk-scoring functions, and the cloud database. Finally, the capstone connection demonstrates how SentinelX uses NLP to analyze cyber threat data, serverless computing to calculate risk scores, and cloud storage and dashboards to provide secure, scalable, and real-time threat intelligence.

Task 2 - Service Model Matrix

Complete the matrices by making technical decisions for your capstone project. Select only the rows that are relevant to your capstone design. If a row is not used, write a short technical reason instead of leaving it blank.

Your answer should show how responsibility shifts between cloud provider and student development team, and how the chosen services support the capstone workflow, API design, storage, security, scalability, and evidence submission.

Task 2A - Service Model Responsibility Matrix

Service Model	Capstone Role / Module	Provider Responsibility	Student Responsibility	Evidence to Submit
IaaS	Virtual machines and networking	Manage servers, storage, networking and virtualization	Configure VMs, install OS, deploy application	VM configuration screenshots
PaaS	Backend API and NLP processing	Manage runtime, middleware and scaling	Develop REST APIs, NLP logic and risk scoring	Backend code and API screenshots
SaaS	Dashboard and collaboration	Maintain software and updates	Use software for monitoring and reporting	Dashboard screenshots
XaaS / Specialized Service	Serverless risk scoring	Execute serverless functions and auto scaling	Develop and deploy serverless functions	Function deployment screenshots

Task 2B - Capstone Cloud Service Decision Matrix

Cloud Layer / Function	Selected Tool or Service	Data / Resource Handled	API or Integration Path	Security / Scaling Note
Client access and UI	React Web Application	User requests	HTTPS + REST API	Secure login using HTTPS
API / backend logic	Node.js + Express	Threat analysis requests	REST API	Input validation and authentication
Authentication and identity	Firebase Authentication / JWT	User credentials	Authentication API	Token-based authentication
Database / structured records	MongoDB Atlas	Threat logs and risk scores	MongoDB Driver	Encrypted storage and backup
File / object storage	AWS S3 / Firebase Storage	Log files and reports	Storage API	Access control
Monitoring / logs / usage metrics	AWS CloudWatch	Server logs and metrics	Monitoring API	Automatic alerts
Analytics / reporting output	React Dashboard	Risk reports and trends	REST API	Scalable visualization

Technical justification: Explain why the selected cloud service pattern fits your capstone project's users, data, API flow, scalability, and security needs.

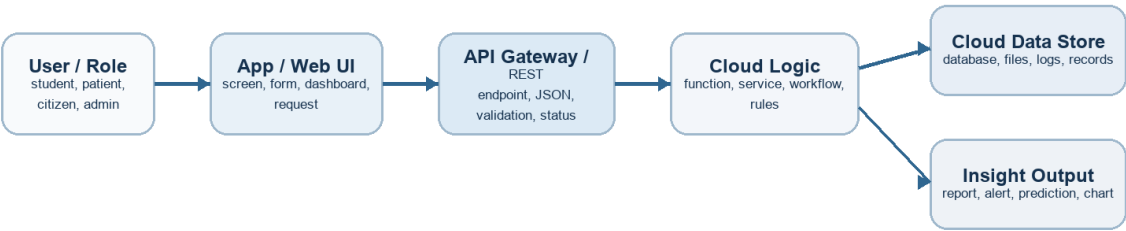
The SentinelX project uses cloud services to process cyber threat data efficiently. REST APIs connect the frontend with the backend, NLP analyzes threat intelligence, serverless functions calculate risk scores automatically, MongoDB Atlas stores data securely, and cloud monitoring provides scalability and reliability.

Task 3 - Capstone Cloud Flow Mapping

Study the sample architecture-flow template below. Redraw it in draw.io / diagrams.net using your own capstone architecture. Replace every generic label with your actual user role, screen, API endpoint, backend logic, cloud storage, analytics output, security control, and scalability or elasticity decision.

Sample Capstone Cloud Flow Template

Replicate this structure using your own capstone screens, APIs, cloud services, data stores, and outputs.



Cross-cutting cloud decisions to label in your own diagram

- Authentication / IAM
Who is allowed to access each API and data item?
- Scalability
Which component must handle growing users or records?
- Elasticity
Which component can expand or shrink based on demand?
- Monitoring
What logs, usage metrics, or errors will be observed?
- Security
Where are validation, privacy, backup, and access control applied?

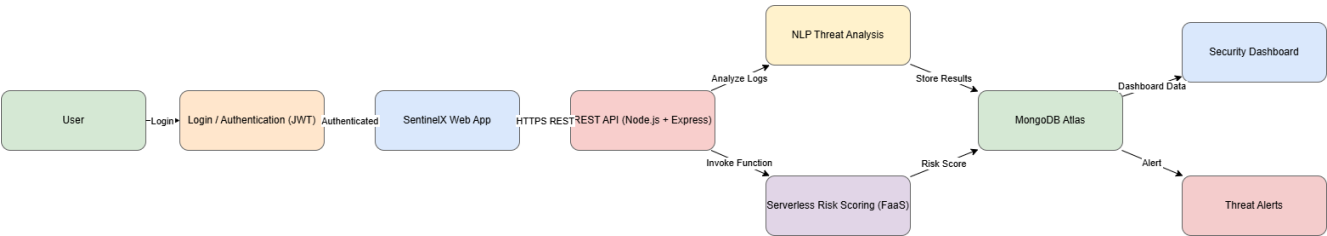
Student instruction: redraw this flow in draw.io/diagrams.net and replace every label with your own capstone architecture.

Student Drawing Requirement

Do not copy the sample as it is. Use it as a layout model, then create your own capstone-specific flow diagram with meaningful labels, arrows, data movement, API names, and cloud service choices.

Flow Component	Student Entry
User role and action	User logs into the SentinelX web application and uploads cyber threat logs for analysis
Frontend / app screen	SentinelX Web Application (React.js)
API endpoint and method	REST API (POST /analyzeThreat) using HTTPS.
Backend processing rule	Node.js + Express receives the request, performs NLP-based threat analysis, invokes Serverless Risk Scoring, and stores the results.
Cloud database / storage operation	MongoDB Atlas stores threat logs, risk scores, and analysis reports securely.
Response, alert, report, or dashboard output	Security Dashboard displays the risk score and generates threat alerts and reports.
Security, scalability, and elasticity labels added in diagram	JWT Authentication, HTTPS Encryption, Auto Scaling, Serverless Computing (FaaS), Elastic Cloud Storage, and Secure Database Access.

Paste your redrawn capstone cloud-flow diagram here.



Brief explanation: explain one complete request-response path from user action to cloud response.

The cloud flow diagram illustrates the working of the SentinelX Cloud Threat Intelligence Using NLP and Serverless Risk Scoring system. A user first logs into the SentinelX web application through a secure authentication mechanism using JWT. The application sends requests to the backend through REST APIs. The backend processes the incoming threat data using an NLP engine to identify malicious patterns and then invokes a serverless risk-scoring function to calculate the threat level. The processed results are stored securely in MongoDB Atlas. Finally, the system displays the results on the Security Dashboard and generates real-time

threat alerts whenever high-risk activities are detected. The cloud architecture supports security, scalability, elasticity, and reliable data processing

Task 4 - Individual Reflection

Answer the following from your own perspective. Avoid team-common wording.

1. What is the strongest cloud concept represented in your capstone project?

The strongest cloud concept in my project is serverless cloud computing. SentinelX uses serverless functions to analyze threats and calculate risk scores without managing servers. This improves efficiency, reduces infrastructure overhead, and allows automatic scaling when the workload increases.

2. Where will elasticity or scalability become important in your capstone product?

Elasticity and scalability become important when many users or threat logs are processed at the same time. Cloud resources can automatically scale up during heavy traffic and scale down when demand decreases, ensuring good performance and cost efficiency

3. Which API or web-service idea is most relevant to your project and why?

REST APIs are the most relevant because they enable secure communication between the web application, NLP engine, serverless functions, and cloud database. They allow different components of SentinelX to exchange data quickly and reliably.

4. What did you personally understand better after preparing this concept map and matrix?

After preparing this assessment, I gained a better understanding of cloud service models, deployment models, APIs, virtualization, and serverless computing. I also learned how these concepts work together to build a secure and scalable cloud-based cybersecurity application

Evaluation Rubric - 100 Marks

Criteria	Marks	Expected Standard
Concept identification	20	Major CO1 concepts are accurately identified: cloud fundamentals, characteristics, evolution, service models, deployment models, virtualization, web services, SOAP/REST, APIs, elasticity, and scalability.
Relationship mapping	20	Links between concepts are accurate, meaningful, and labeled. The map should explain how concepts depend on, enable, consume, provide, scale, or control one another.
Hierarchy and organization	15	Concepts are arranged in a logical hierarchy with clear grouping, readable flow, and no disconnected keyword clusters.
Technical relevance	15	The map strongly reflects cloud course content and avoids decorative or generic non-technical content.
Service model matrix	15	IaaS, PaaS, SaaS, and XaaS are compared with provider responsibility, user responsibility, examples, capstone usage, and risks or limitations.
Capstone alignment and individual reflection	10	The student's own capstone interpretation is visible in the concept map, flow mapping, justification, and reflection. Team-common answers are avoided.
Visual clarity and digital evidence	5	Diagram is neat, readable, exported properly, and supported with GitHub evidence including source and exported files.

Faculty Evaluation Record

Evaluator Name		Marks Awarded	
Observation		Signature	